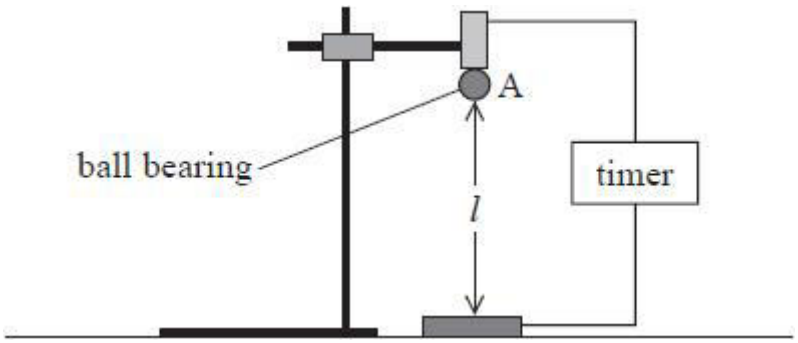
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Questions

Q1.

A student carried out an experiment to measure the acceleration of free-fall *g*.

A ball bearing was released from position A. The time *t* for the ball bearing to fall a distance *l* was measured. This was repeated for decreasing values of *l*.



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Explain how the measurements obtained can be used to determine a value for *g* in m s^{-2}.
Your answer should include a sketch on the axes below of the graph that the student would expect to obtain.

(5)



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(Total for question = 5 marks)

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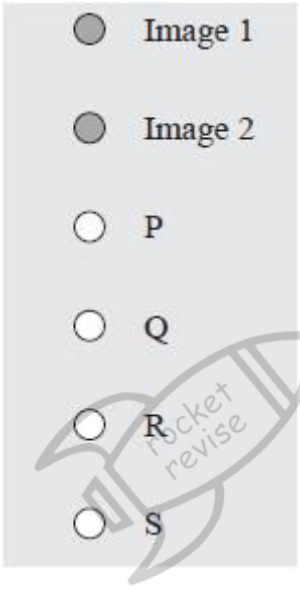
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Q2.

A student used a falling sphere to determine the acceleration of free fall.

A camera produced images of the sphere at constant time intervals as it fell.

The positions of the sphere in the first two images are shown. Image 1 shows the sphere's position at the instant it was released.



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Which of the positions P, Q, R or S will the sphere be at in Image 3?

(1)

- **A** P
- **B** Q
- **C** R
- **D** S

(Total for question = 1 mark)

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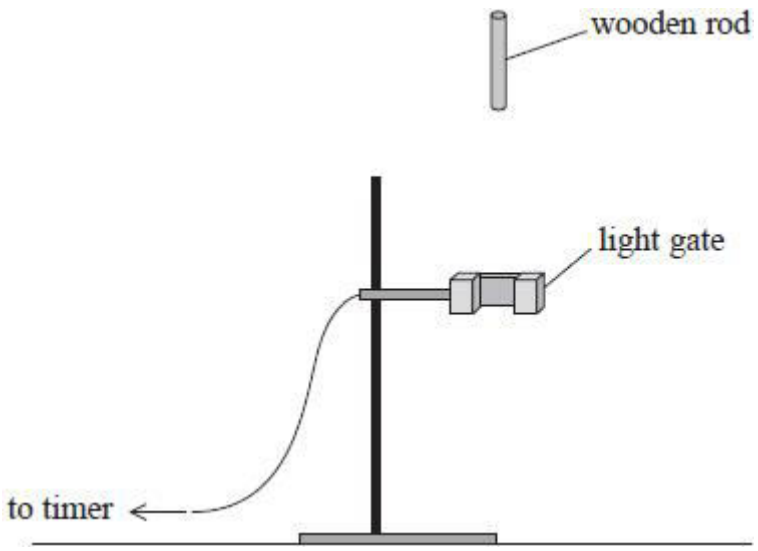
Q3.

A student carries out an experiment to determine a value for g, the acceleration of free fall. A short wooden rod is released above a light gate. A timer connected to the light gate is used to measure the time taken for the wooden rod to pass through the light gate.

The experimental arrangement is shown.

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The student uses the equation $v^2 = u^2 + 2as$, where $u = 0$, and a graphical method to determine a value for g.

(a) State the additional measurements the student should take.

(2)

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(b) Describe how the velocity v of the wooden rod as it passes through the light gate can be determined accurately.

(2)

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(c) Describe how the student can determine a value for g using a graphical method.

(3)

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(Total for question = 7 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A student used a falling sphere to determine the acceleration of free-fall *g*. The sphere was released from rest.

Which two quantities would require the fewest measurements to be taken in order to determine *g*?

☐ **A** displacement and initial velocity

☐ **B** displacement and time

☐ **C** final velocity and displacement

☐ **D** final velocity and time

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(Total for question = 1 mark)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A student determined the acceleration of free fall *g*.

She measured the time *t* for a ball bearing to fall from rest through a distance *s*.

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She repeated this for a range of values of *s*.

Which row of the table shows the graph that would give a gradient equivalent to *g*?

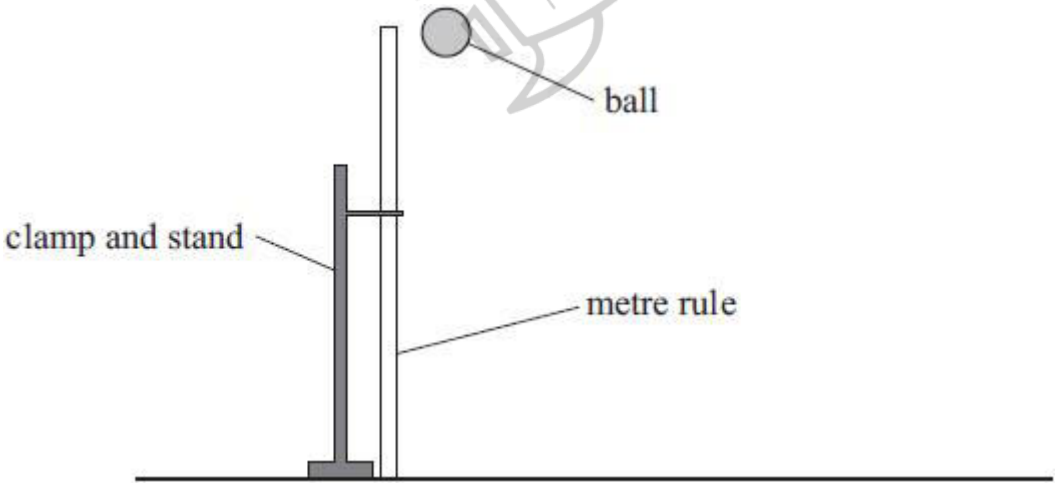
	y-axis	*x*-axis
☒ A	*s*	$2t^2$
☐ B	$2s$	t^2
☐ C	*s*	$2t$
☐ D	$2s$	*t*

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(Total for question = 1 mark)

Q6.

A student clamped a metre rule so that it was vertical. She dropped a ball from rest near the top of the metre rule, as shown.



A strobe emits flashes of light. The time interval between flashes is constant.

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The student photographed the falling ball using strobe lighting. The ball can be seen at different heights in the photograph, as shown.

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(Source: © sciencephotos/Alamy Stock Photo)

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For each flash of light, the student determined the distance fallen by the ball.

(a) She took one photograph using a strobe app on a mobile phone.

She took a second photograph using a laboratory strobe.
The time interval between flashes was the same for the strobe app and for the laboratory strobe. Each flash of light from the laboratory strobe has a smaller duration than each flash from the mobile phone.
Explain how the smaller duration of each flash from the laboratory strobe affected the uncertainty in the measurement of the distance fallen.

(2)

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(b) The student recorded the distances s fallen by the ball and corresponding values of the time t.

(i) Explain why a graph of s against t^2 gives a straight line.

(2)

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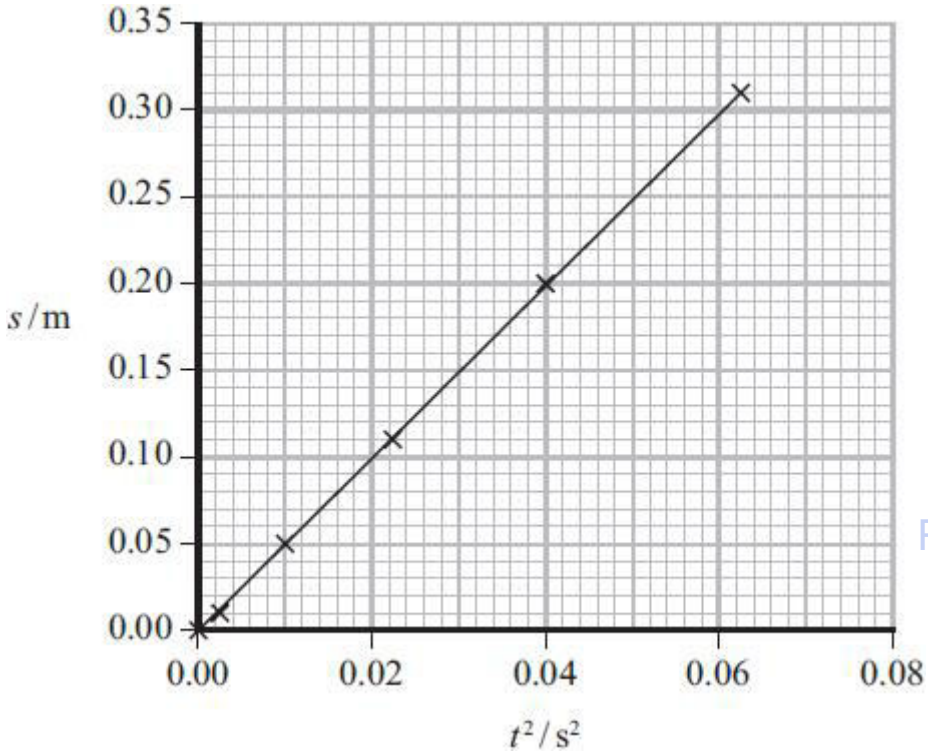
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(ii) The student plotted a graph of s against t^2, as shown.

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Determine the acceleration of free fall, *g*, using the student's graph.

(2)

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g = ...

(Total for question = 6 marks)

Q7.

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A student measures the time *t* taken for a ball bearing to fall different measured distances *s* from rest. The student uses his measurements to plot a graph with a gradient equal to the acceleration due to gravity *g*.

Which row of the table shows a graph with a gradient equal to *g*?

	y-axis	*x*-axis
☐ **A**	s	t^2
☐ **B**	$2s$	t^2
☐ **C**	t^2	s
☐ **D**	t^2	$2s$

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(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A ball bearing falls vertically from rest through a distance of 50 cm in a time of 0.32 s.

Which expression gives the acceleration of the ball bearing in m s^{-2}?

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- ☐ **A** $1 \div 0.32^2$
- ☐ **B** $0.5 \div 0.32$
- ☐ **C** $100 \div 0.32^2$
- ☐ **D** $50 \div 0.32$

(Total for question = 1 mark)

Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A student measures the time taken for a steel ball bearing to fall a measured distance in air. The student uses these measurements to determine the acceleration due to gravity.

Which of the following conditions is needed to determine an accurate value for the acceleration due to gravity?

- ☐ **A** The air flow around the ball bearing should be laminar.
- ☐ **B** The time taken to achieve terminal velocity should be short.
- ☐ **C** The ball bearing should begin to fall before the timer is started.
- ☐ **D** Air resistance and upthrust should both be negligible.

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(Total for question = 1 mark)

Q10.

Answer the question with a cross in the box you think is correct (☒). If you change your mind, put a line through the box (☒) and then mark your new answer with a cross (☒).

A student determined a value for *g* by dropping a metal sphere from rest. The student measured the distance fallen and the time taken for the sphere to fall.

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The student obtained a value for g of 11.2 m s^{-2}.

Which of the following could explain the difference between the student's value and the accepted value?

☒ **A** The air resistance was not negligible.

☒ **B** The student's measured distance was greater than the actual distance.

☒ **C** The sphere did not fall vertically.

☒ **D** The student's measured time was greater than the actual time taken.

(Total for question = 1 mark)

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Q11.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A student is determining the acceleration of free-fall, g.

The student uses a stopwatch to measure the time taken for a ball bearing to fall a measured distance.

The student calculates a value of g that is greater than 9.81 m s^{-2}.

Which of the following could be an explanation for the student's result?

☒ **A** The ball bearing is moving downwards when the student starts the stopwatch.

☒ **B** The ball bearing is moving upwards when the student starts the stopwatch.

☒ **C** The student starts the stopwatch before the ball bearing is dropped.

☒ **D** The student stops the stopwatch after the ball bearing lands.

(Total for question = 1 mark)

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Q12.

Select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An object on the Moon falls a vertical distance of 0.32 m, from rest, in a time of 0.63 s.

Which of the following expressions gives the acceleration due to gravity on the Moon in m s^{-2}?

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☐ **A** $\frac{0.32}{2 \times 0.63}$

☐ **B** $\frac{0.32}{2 \times 0.63^2}$

☐ **C** $\frac{2 \times 0.32}{0.63^2}$

☐ **D** $\frac{2 \times 0.32}{0.63}$

(Total for question = 1 mark)

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Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A student drops a ball and measures the time taken for the ball to reach the ground.
He repeats this for different heights.

He uses a graphical method to determine a value for *g*, the acceleration of free fall.

He plots height on the *y*-axis of his graph.

Which row of the table is correct for the student's graph?

	Quantity plotted on *x*-axis	Gradient of graph
☐ **A**	time	$\frac{1}{g}$
☐ **B**	time	g
☐ **C**	$(\text{time})^2$	$\frac{g}{2}$
☐ **D**	$(\text{time})^2$	$2g$

(Total for question = 1 mark)

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